

HEMANTH AMBATI

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SUMMARY

Software Engineer with a strong foundation in backend development, test automation, and AI/ML integration, skilled in C#, Python, Java, and cloud platforms like Azure. Experienced in designing scalable RESTful APIs and implementing test frameworks to ensure code quality and reliability. Proven ability to build intelligent tools using NLP and generative AI, optimize performance for critical endpoints, and streamline deployment through CI/CD pipelines. Passionate about delivering high-quality, intelligent software solutions that drive engineering excellence and improve development workflows.

TECHNICAL SKILLS

- **Programming Languages:** Python, C#, C++, Java, SQL
- **AI/ML & Data Science:** Generative AI, NLP, Deep Learning, Scikit-learn, PyTorch, TensorFlow
- **Software Development:** Data Structures & Algorithms, Object-Oriented Programming, RESTful APIs, Microservices, Design Patterns, System Design
- **Testing & Quality Engineering:** Test Automation, Selenium, PyTest, JUnit, TestNG, API Testing, Postman, Unit & Integration Testing, Performance Testing, Test Coverage Analysis, CI/CD Integration, Azure DevOps
- **Cloud & DevOps:** Azure AI Services, Azure, Model Deployment, Azure CI/CD
- **Version Control & Collaboration:** GitBash, GitHub, Azure DevOps

PROFESSIONAL EXPERIENCE

BECU

Jun 2025 - Sep 2025

Developer Intern (AI/ML & Full Stack)

- Developed an AI-powered diagnostic agent with Azure OpenAI, MCP, and REST APIs, streamlining Application Insights logs interpretation and cutting resolution time by up to 90 minutes.
- Optimized critical back-end API endpoints by enhancing validation and parallelizing API calls, reducing response times by 60% and increasing user satisfaction
- Delivered real-time execution dashboards with Allure and integrated reporting systems, increasing collaboration and traceability across teams.
- Implemented CI/CD pipelines in Azure DevOps to streamline deployment and improve delivery efficiency.

University of Louisville

Aug 2023 - Jan 2025

Machine Learning Engineer (Assistant)

- Built and optimized machine learning pipelines for student projects using Python, Pandas, NumPy, and Scikit-learn, improving model performance and workflow efficiency.
- Developed end-to-end ML solutions, including data preprocessing, feature engineering, model training, and evaluation, resulting in a 20% increase in predictive accuracy for advanced analytics projects.
- Debugged classification and regression models, enhancing reliability and reducing error rates for deployed academic tools.
- Conducted hands-on workshops on ML fundamentals and supervised students implementing predictive analytics and NLP-based applications, resulting in higher student engagement and successful completion of real-world projects.
- Leveraged version control (GitHub) and collaborative workflows to streamline experimentation and ensure reproducible results.

IBM

May 2023 - Jun 2023

AI intern

- Developed an AI-powered document classification system using Python, Scikit-learn, and NLP techniques.
- Automated information extraction from unstructured documents, improving data retrieval efficiency by 30%.
- Conducted experiments to evaluate AI models, optimizing accuracy and performance for large-scale text datasets.

Wipro Technologies

Mar 2022 - Jul 2022

Java Full Stack Developer Intern

- Designed and implemented a robust distributed system with microservices and RESTful APIs, greatly enhancing system scalability, reliability, and performance.
- Enhanced database query performance by 40% through indexing and caching.
- Enhanced cloud service deployment automation with Bash scripting, cutting deployment time by 30% and minimizing manual errors for increased efficiency.

Microsoft

Nov 2021 - Feb 2022

Future Ready Talent Intern

- Built an Azure-based chatbot using NLP techniques, automating responses for customer inquiries.
- Developed a RESTful API for AI-powered document retrieval, reducing manual effort in text-based queries.
- Integrated AI-driven automation into cloud services, optimizing data processing workflows.

EDUCATION

University of Louisville

Master of Science, Computer Science

Aug 2023 - May 2025

PROJECTS

Application Insights Agent using Azure LLM and MCP.

Aug 2025 - Sep 2025

- Developed a secure AI-powered diagnostic agent for Application Insights using Azure OpenAI, Model Context Protocol (MCP), and REST APIs.
- Automated prompt-to-KQL query generation, log retrieval, and summarization, reducing manual debugging effort.
- Cut incident resolution time by 60–90 minutes per issue by delivering automated root cause analysis and recommended fixes.
- Enhanced engineering productivity by streamlining monitoring workflows and eliminating manual dashboard navigation.

Emotion Based music recommendation system.

Sep 2021 - Mar 2022

- Designed an AI-driven recommendation system that analyzes facial expressions to suggest personalized music.
- Implemented sensor fusion techniques using OpenCV & TensorFlow for real-time facial recognition and emotion detection.
- Optimized model accuracy by 20% using data augmentation and transfer learning techniques.

FAQ College Chatbot.

Dec 2021 - Feb 2022

- Developed an NLP-based chatbot using Azure AI, Scikit-learn, and Python to automate university FAQ responses.
- Designed a scalable multi-threaded backend capable of handling concurrent user queries efficiently.
- Integrated Named Entity Recognition (NER) and sentiment analysis, improving chatbot response accuracy by 30%.

CERTIFICATIONS

- **Microsoft Certified: Azure AI Fundamentals**
- **Microsoft Certified: Azure Fundamentals**
- **Programming in HTML5 with JavaScript and CSS3:**Great Learning
- **Introduction to Programming Using Python:**Internshala
- **Introduction to Cybersecurity:**Cisco